AMLODIPINE

Once daily

Ankle oedema common

CHEMICAL NAME 3-ethyl 5-methyl (4RS)-2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-6-methyl-1,4-dihydropyridine-3,5-dicarboxylate (besilate)

CLASS Dihydropyridine calcium-channel blocker (long-acting)

TRADE NAMES Norvasc · Istin · Amlor · Exforge (with valsartan)

THERAPEUTIC USES / INDICATIONS

- Hypertension (first-line, especially elderly and Black patients).
- Chronic stable angina and vasospastic (Prinzmetal) angina.
- Raynaud's phenomenon (off-label).

ADMINISTRATION

- **PO** once daily at the same time, with or without food.
- Gradual onset — no reflex tachycardia.

DOSAGES

- **Adult:** 5 mg once daily; max 10 mg.
- Elderly/hepatic impairment: start 2.5 mg.
- **Children 6–17 yr:** 2.5–5 mg once daily.

PHARMACOKINETICS

- **Onset:** 6–12 h; full effect 1–2 weeks. **Duration:** 24 h+.
- Bioavailability 60–65%; metabolised in liver (CYP3A4).
- Excreted in urine as metabolites. **t½:** 30–50 h.

PHARMACODYNAMICS

- Blocks **L-type calcium channels** in vascular smooth muscle → arteriolar vasodilation → ↓ peripheral resistance and BP.
- Coronary vasodilation; minimal effect on heart rate or conduction.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Ankle/peripheral oedema (dose-related, not due to fluid overload).
- Flushing, headache, dizziness.
- Palpitations, fatigue.

ADVERSE EFFECTS (SERIOUS)

- Gingival hyperplasia.
- Hypotension.
- Rarely worsening angina at start.
- Hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Cardiogenic shock; severe aortic stenosis.
- Unstable angina (except vasospastic).
- Severe hypotension.
- **Caution:** severe hepatic impairment, heart failure.

KEY INTERACTIONS

- **Simvastatin:** limit to 20 mg/day.
- **CYP3A4 inhibitors** (clarithromycin, itraconazole): ↑ amlodipine.
- **Ciclosporin, tacrolimus:** ↑ levels.
- **Grapefruit juice:** minor ↑ levels.

NURSING MANAGEMENT

- Monitor BP and HR.
- Check ankles for oedema; distinguish from heart failure (no weight gain/breathlessness).
- Assess for headache, flushing.
- Encourage dental hygiene (gum overgrowth).

PATIENT EDUCATION

- Take once daily even when you feel well.
- Ankle swelling may occur — elevate legs; report if severe.
- Brush and floss regularly.
- Rise slowly.

KEY POINTS

- Long-acting DHP CCB; once daily.
- Ankle oedema common.
- First-line for hypertension.
- Limit simvastatin to 20 mg.

SOURCES CHECKED **[313]** Amlodipine / Valsartan 10 mg / 160 mg film-coated tablets (UK SmPC) **[314]** Amlodipine and Atorvastatin, oral (US label) **[315]** Amlodipine and atorvastatin, oral (US label)

Full references p. 432–457